

■ 1.9.7 Special Topic: External Functions

On many computer systems, *Mathematica* can not only execute complete external programs, but can also call individual functions inside appropriately-prepared external programs. In this way, you can set up an external program which contains a “library” of functions that can be accessed from *Mathematica*.

This kind of communication with external programs is again done through pipes. Packets of data, specifying the function to call, and the arguments to give, are sent from *Mathematica* through the pipe to the external program. Routines inside the external program then decode the packet, call the appropriate function, and then return the result to *Mathematica*. Versions of the routines that are needed for this decoding are usually distributed with *Mathematica* in source code form.

<code>StartProcess["command"]</code>	execute an external program, and set up <i>Mathematica</i> definitions for the functions it contains
<code>CallProcess["command", f, {arg₁, arg₂, ...}]</code>	call the function <i>f</i> in an external process
<code>EndProcess["command"]</code>	stop an external process

Calling functions in external processes.

The same basic mechanism that allows *Mathematica* to call functions inside an external program can also be used to allow an external program to call *Mathematica*. In this way, you can effectively treat *Mathematica* as a “library” that your external programs can call on.